SIAMIS ICT Weekly Lesson Plan

Teacher: Ray Joshua Balingkit | Year 10–11 | Term 2 | Week 5

Year 10 course sequence: Units 1–3. Two 45-minute lessons. Year 11 progression: Units 4–6.

Lesson overview

This week covers network logins, passwords and firewalls and encryption, wireless security, VPNs, permissions and logs.

Pearson Edexcel International GCSE (9-1) ICT Student Book, printed pages 90-93

End of week assessment

Check accurate terminology, explanations and application through the relevant book activities and questions. Record misconceptions and use the next feedback or consolidation session for corrections. Assess only content already taught.

Lesson 1

Network logins, passwords and firewalls

Date: 1 February 2027 | Duration: 45 minutes

Learning objective

By the end of this lesson, students will be able to explain and apply network logins, passwords and firewalls, using accurate textbook terminology and a justified example.

Lesson flow

Hook 5 minutes

Use a short scenario, image or retrieval question to introduce network logins, passwords and firewalls.

Introduce 8 minutes

Teach network logins, passwords and firewalls from printed pages 90-93; include the relevant definitions, diagrams and examples.

We do 10 minutes

Work through a textbook example or comparison for network logins, passwords and firewalls; explain each decision and check understanding.

You do 17 minutes

Complete the relevant textbook activities and questions for network logins, passwords and firewalls. Use a practical demonstration or scenario where appropriate; finish remaining practice as homework.

Check 5 minutes

Explain one key idea about network logins, passwords and firewalls and justify an answer; record misconceptions for the next lesson.

Differentiation

Support: Provide labelled diagrams, vocabulary, a structured response frame and a worked example. Apply agreed access arrangements during independent assessment without supplying answers.

Extension: Justify an alternative solution, explain a limitation or apply the same concept to a new scenario using evidence.

Lesson 2

Encryption, wireless security, VPNs, permissions and logs

Date: 3 February 2027 | Duration: 45 minutes

Learning objective

By the end of this lesson, students will be able to explain and apply encryption, wireless security, VPNs, permissions and logs, using accurate textbook terminology and a justified example.

Lesson flow

Hook 5 minutes

Use a short scenario, image or retrieval question to introduce encryption, wireless security, VPNs, permissions and logs.

Introduce 8 minutes

Teach encryption, wireless security, VPNs, permissions and logs from printed pages 90-93; include the relevant definitions, diagrams and examples.

We do 10 minutes

Work through a textbook example or comparison for encryption, wireless security, VPNs, permissions and logs; explain each decision and check understanding.

You do 17 minutes

Complete the relevant textbook activities and questions for encryption, wireless security, VPNs, permissions and logs. Use a practical demonstration or scenario where appropriate; finish remaining practice as homework.

Check 5 minutes

Explain one key idea about encryption, wireless security, VPNs, permissions and logs and justify an answer; record misconceptions for the next lesson.

Differentiation

Support: Provide labelled diagrams, vocabulary, a structured response frame and a worked example. Apply agreed access arrangements during independent assessment without supplying answers.

Extension: Justify an alternative solution, explain a limitation or apply the same concept to a new scenario using evidence.

Materials

Teacher computer and display

School-approved applications, class save folder and fictional scenarios

Relevant textbook diagrams, activities and chapter questions

Student notebooks and a teacher assessment checklist

Teacher guidance: explain historical wireless standards as textbook examples; do not recommend obsolete security settings or change managed network settings.

Teacher reflection

What worked well? _____

What needs reteaching or adjustment? _____